Voltage: energy per charge

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Interpret potential difference as energy transferred per unit charge.

SUCCESS CRITERIA - I can...

1. Use volts as joules per coulomb.
2. Calculate energy transfer with E=qV.
3. Distinguish voltage from current.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “voltage” and “charge” to explain your answer. Sentence starters are optional.

voltage: electric potential difference; energy transferred per charge

charge: a property responsible for electric interactions

energy: a quantity tracked through storage and transfers in a system

01 Warm-up: bring back an earlier idea

Two objects have charges **-3 nC** and **-5 nC**. **Predict attraction or repulsion** and **draw both force directions**.

Sentence starter: I predict … . I used … to make this prediction because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: energy transferred = qV; V = energy transferred/q. Magnitudes here; **1 V** = **1 J/C**. Voltage is a difference between two points.

Worked reasoning: Moving **2 C** through a potential difference of **3 V** changes electric potential energy by q ΔV=**6 J**. Voltage is energy change per charge, not energy by itself.

Your turn: Draw two labeled potential levels and an energy account for a positive charge moved between them.

Sentence starter: In my work, I used … to show … . I labeled it … .

Initial / supply _____ Final / component _____

V: ____ J/C V: ____ J/C

Charge: ____ C Energy transfer: ____ J

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Compare a 3 V** and a **6 V** battery for the same amount of charge transferred.
2. Which supplies more energy per coulomb?

Sentence starter: Both … . They differ in … because … .

Initial / supply _____ Final / component _____

V: ____ J/C V: ____ J/C

Charge: ____ C Energy transfer: ____ J

Marzano check - circle: ① ② ③

04 Apply the physics

A **6 V** battery moves **2 C** through a small classroom car circuit.

1. **Find the energy transferred**.
2. **Label voltage as energy per** coulomb and **show the total transfer** on the diagram.

Sentence starter: I found … [unit]. This value means … in this situation.

Initial / supply _____ Final / component _____

V: ____ J/C V: ____ J/C

Charge: ____ C Energy transfer: ____ J

Marzano check - circle: ① ② ③

Voltage: energy per charge

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student says a **9 V** battery contains **9** coulombs. **Explain the units** and why voltage does not state total battery capacity.

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A wall is **60 m** away. An echo returns after **0.6 s**. A student divides one-way distance by this time. **Find the correct speed** and **explain the student's error**. **Label the full travel path** and the recorded time.

Word help: echo = a sound received after reflection

Sentence starter: The step I would change is … . I would change it because I found … .



Marzano check - circle: 1 2 3

07 Exit check

What does a potential difference of **4 V** mean in J/C?

Sentence starter: Both … . They differ in … because … .

Initial / supply	Final / component
V: ____ J/C	V: ____ J/C

Charge: ____ C Energy transfer: ____ J

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Two devices use the same **6 V** supply but move charges **1** and **3 C**. **Compare transferred energies**.

Sentence starter: Both … . They differ in … because … .

Initial / supply	Final / component
V: ____ J/C	V: ____ J/C

Charge: ____ C Energy transfer: ____ J

Marzano check - circle: 1 2 3